HW 2

1. Two-Wavelength Sodium Doublet Theory (12 pts) In this question, you'll consider a Michelson interferometer illuminated by a sodium lamp, modeled as a source composed of two single-wavelength spectral lines located at $\lambda_1 = 589.6\,\text{nm}$ and $\lambda_2 = 589.0\,\text{nm}$.

1. As in class, assume path lengths d_1 and d_2 from source to detector via the two arms and let the source be stable over time intervals greater than $t_1 = d_1/c$ or $t_2 = d_2/c$. Neglect polarization. Assume what's seen at the detector is one (attenuated) copy of the incoming time-dependent wave, $E(t)/2$, minus another copy shifted in time by $\tau \equiv t_2 - t_1$. Explain why $\tau = 2(\ell_2 - \ell_1)/c$, where ℓ_1 (or ℓ_2) is the length of interferometer arm 1 (or 2).

τ is the difference in phase shift from each of the two paths. Assuming that overall phase is irrelevant, the total shift τ can be put onto one path. The difference in length, then, is what matters. The wave travels an extra distance $\ell_2 - \ell_1$ at a speed c on one way down the arm. The beams have to travel the arm lengths twice - going out to the mirror and back, so it takes time $\frac{2(l_2 - l_1)}{c}$ extra to travel the second path.

2. Let $E(t) = E_1^{(+)}e^{-i\omega_1 t} + E_2^{(+)}e^{-i\omega_2 t} + c.c.$, so the source consists of two angular frequencies ω_1 and ω_2. Eventually we'll set the amplitudes of the two plane waves equal to each other. Calculate the intensity at the detector as a function of τ. Don't obsess over overall constant factors—only the *form* of the intensity as a function of τ matters. Hint: Start with the general result for the interferometer's output intensity as found in Lecture 8. Within this calculation, once you multiply out all of the terms, the time average should get rid of any terms that oscillate (as functions of time) with many cycles per detection time. Since detection times are at least 10^{-9} seconds, even terms like $e^{\pm i(\omega_2-\omega_1)t}$ are fast-oscillating, assuming $\omega_2 - \omega_1 \gg 10^{10}/\text{s}$, which for optical wavelengths translates roughly to the condition $\lambda_1 - \lambda_2 \gg 10^{-3}\,\text{nm}$. Only terms involving only τ and not t remain! Remember, τ is not a time interval in the lab procedure, but something you control by setting the interferometer to a particular path length difference.

We know that

$$\tau = \frac{2(l_1 - l_2)}{c}$$

The phase shift is $\omega\tau$.

$$I_{OUT} = \frac{1}{2}I_{IN} - \frac{1}{\mu_0 c}\mathrm{Re}\langle E^{(+)}(t+\tau)E^{(-)}(t)\rangle = \frac{1}{2}I_{IN}[1 - \mathrm{Re}\,g^{(1)}(\tau)]$$

$$\text{where } g^{(1)}(\tau) \equiv \frac{\langle E^{(+)}(t+\tau)E^{(-)}(t)\rangle}{\langle E^{(+)}(t)E^{(-)}(t)\rangle}$$

$$I_{\text{out}} = \frac{1}{2}I_{in}[1 - Reg^{(1)}(t)]$$

We can calculate the coherence function for this

$$\left\langle E^{(+)}(t)E^{(-)}(t)\right\rangle = E^{(+)}E^{(-)} = E_1^2 + E_2^2$$

We also have

$$\left\langle E^{(+)}(t+\tau)E^{(-)}(t)\right\rangle = (E_1^{(+)}E_1^{(-)}e^{-i\omega_1\tau} + E_2^{(+)}E_2^{(-)}e^{-i\omega_2\tau})$$

This gives

$$g^{(1)}(\tau) = \frac{(E_1^{(+)}E_1^{(-)}e^{-i\omega_1\tau} + E_2^{(+)}E_2^{(-)}e^{-i\omega_2\tau})}{E_1^2 + E_2^2}$$

We can find

$$Re(g^{(1)}(\tau)) = \frac{E_1^{(+)}E_1^{(-)}\cos(\omega_1\tau) + E_2^{(+)}E_2^{(-)}\cos(\omega_2\tau)}{E_1^2 + E_2^2}$$

We can also find

$$
\begin{aligned}
I_{\text{in}} &= \frac{2}{\mu_0 c}\left\langle E^+(t)E^-(t)\right\rangle \\
&= \frac{2}{\mu_0 c}\left\langle E_1^+ e^{-i\omega_1 t}E_1^- e^{i\omega_1 t}E_1^+ + E_2 e^{-i\omega_2 t}E_2^- e^{i\omega_2 t} + E_1^+ e^{-i\omega_1 t}E_2^- e^{+i\omega_2 t} + E_2^+ e^{-i\omega_2 t}E_1^- e^{+i\omega_1 t}\right\rangle \\
&= \frac{2}{\mu_0 c}\left\langle E_1^+E_1^- + E_2^+E_2^- + E_1^+E_2^- e^{i(\omega_2-\omega_1)t} + E_1^-E_2^+ e^{i(\omega_1-\omega_2)t}\right\rangle
\end{aligned}
$$

Assuming that $\omega_2 - \omega_1$ is large enough, those time varying terms die. Thus,

$$
I_{\text{in}} = \frac{2}{\mu_0 c}\left\langle E_1^+E_1^- + E_2^+E_2^-\right\rangle
$$

We therefore have

$$
\begin{aligned}
I_{\text{out}} &= \frac{1}{\mu_0 c}(E_1^+E_1^- + E_2^+E_2^-)\left(1 - \frac{E_1^{(+)}E_1^{(-)}\cos(\omega_1\tau) + E_2^{(+)}E_2^{(-)}\cos(\omega_2\tau)}{E_1^2+E_2^2}\right) \\
&= \frac{1}{\mu_0 c}\left((E_1^+E_1^- + E_2^+E_2^-) - E_1^{(+)}E_1^{(-)}\cos(\omega_1\tau) - E_2^{(+)}E_2^{(-)}\cos(\omega_2\tau)\right)
\end{aligned}
$$

3. Simplify to demonstrate that you have three terms in the expression for output intensity: a constant plus a cosine at each frequency. Except for the constant, note that the Fourier transform of your answer (with τ now treated as the time variable) contains spikes at the two frequencies of the spectral lines with amplitudes proportional to their intensities. This generalizes; the Fourier transform of the interference pattern is the spectrum.

We found previously that

$$
\begin{aligned}
I_{\text{out}} &= \frac{1}{\mu_0 c}\left((E_1^+E_1^- + E_2^+E_2^-) - E_1^{(+)}E_1^{(-)}\cos(\omega_1\tau) - E_2^{(+)}E_2^{(-)}\cos(\omega_2\tau)\right) \\
&= \frac{1}{\mu_0 c}(E_1^2 + E_2^2 - E_1^2\cos\omega_1\tau - E_2^2\cos\omega_2\tau)
\end{aligned}
$$

This is a constant $E_1^2 + E_2^2$ minus two cosine terms for each wave.
Combined, it gets the funkier but maybe nicer

$$
I_{\text{out}} = \frac{1}{\mu_0 c}(E_1^2(1-\cos\omega_1\tau) + E_2^2(1-\cos\omega_2\tau))
$$

4. Assume equal amplitudes for the two frequency components of the sodium doublet. Write an expression for $\frac{I_{OUT}(\tau)}{I_{IN}}$. Use trig identities to simplify your answer in terms of the average angular frequency $\omega_0 = \frac{\omega_1+\omega_2}{2}$ and the difference $\Delta\omega = \omega_1 - \omega_2$.

$$
I_{\text{out}} = \frac{1}{2}I_{in}[1 - Re g^{(1)}(t)]
$$

so

$$
\frac{I_{\text{out}}}{I_{\text{in}}} = \frac{1}{2}[1 - Re(g^1(\tau))]
$$

This is

$$
Re(g^{(1)}(\tau)) = \frac{E_1^{(+)}E_1^{(-)}\cos(\omega_1\tau) + E_2^{(+)}E_2^{(-)}\cos(\omega_2\tau)}{E_1^2+E_2^2}
$$

$$
\frac{I_{\text{out}}}{I_{\text{in}}} = \frac{1}{2}\left[1 - \frac{E_1^2\cos(\omega_1\tau) - E_2^2\cos(\omega_2\tau)}{E_1^2+E_2^2}\right]
$$

If we assume that $E_1^2 = E_2^2$, then we get the nice trig identity.

$$
\cos\theta + \cos\phi = 2\cos\frac{1}{2}(\theta+\phi)\cos\frac{1}{2}(\theta-\phi)
$$

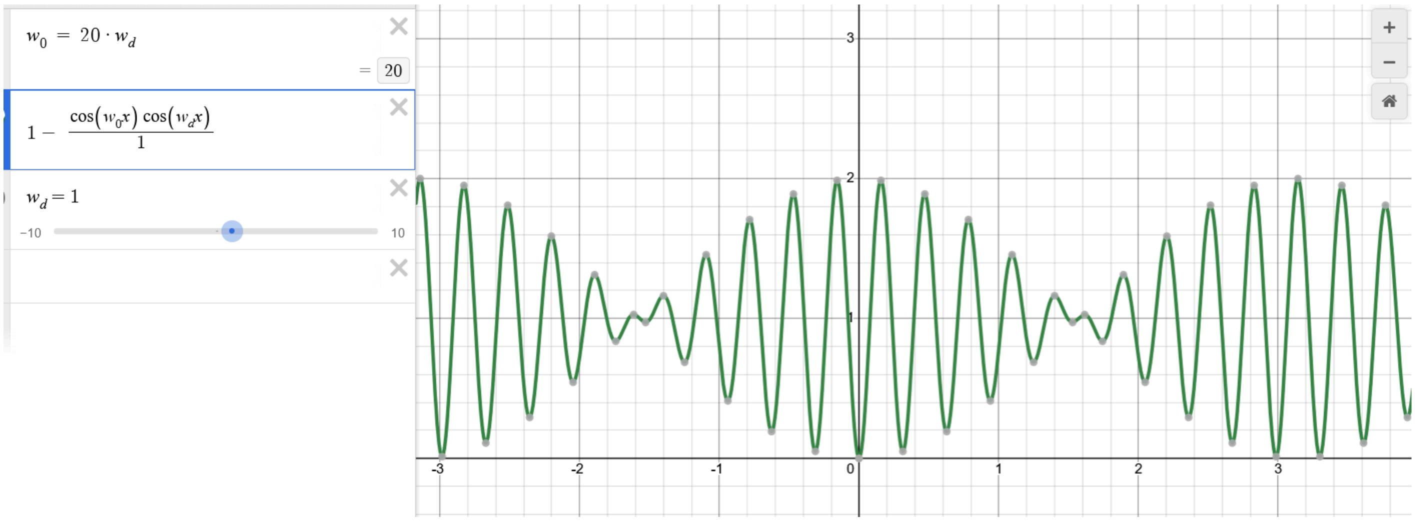
We can relabel this in the context of our function as

$$2 \cos \omega_0 \tau \cos \Delta\omega\tau$$

We get

$$\frac{I_{\text{ out}}}{I_{\text{ in}}} = 1 - \frac{\cos \omega\tau \cos \Delta\omega\tau}{2E_0^2}$$

5. Plot $\frac{I_{OUT}(\tau)}{I_{IN}}$ for the case where $\omega_0 \approx 20\Delta\omega$, which is not nearly as big a factor as for the sodium doublet, but produces a reasonable plot.



6. Your plot should show "fast fringes" modulated by a slower envelope. The "fast fringes" go to from maximum to maximum as the movable mirror is translated over a distance that is on the scale of a single wavelength. The envelope goes from "no fringes" to "bright fringes" and back to "no fringes" on much longer distance scales. Find the amount that $\ell_2 - \ell_1$ must change to go from "no fringes" back to "no fringes". Express your answer in terms of the average wavelength and the separation in wavelengths of the doublet. Then calculate it as a numerical distance.

$\lambda_1 = 589.6\,\text{nm}$ and $\lambda_2 = 589.0\,\text{nm}$.

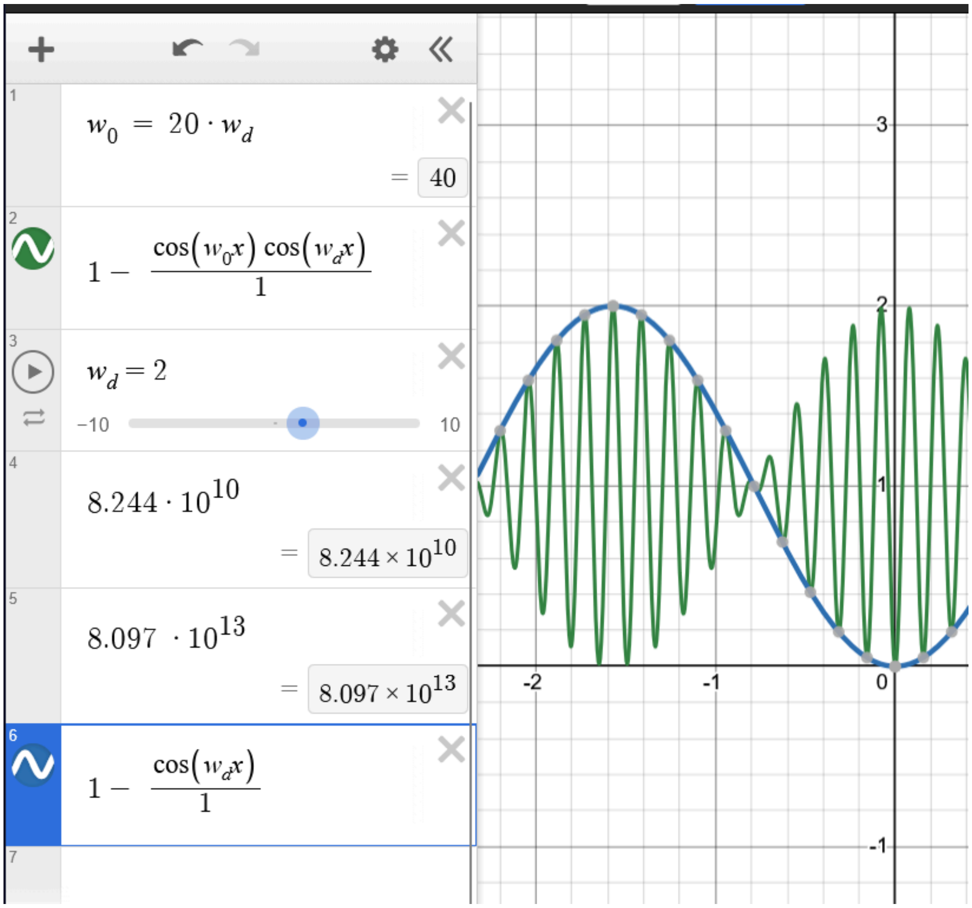
We know that

$$\begin{aligned} 2\pi\omega\lambda &= c \\ \omega_1 &= \frac{3e^8 \text{ m}}{2\pi \cdot 589.6 \text{ s } \cdot \text{ nm}} \\ &= 8.098 * 10^{13} \\ \omega_1 &= \frac{3e^8 \text{ m}}{2\pi \cdot 589 \text{ s } \cdot \text{ nm}} \\ &= 8.106 * 10^{13} \end{aligned}$$

This gives us

$$\begin{aligned} \Delta\omega &= -8.249 * 10^{10} \\ \omega_0 &= 8.098 * 10^{13} \end{aligned}$$

One may note that the modulation of the signal follows $\Delta\omega$



Thus the full period is crossed when

$$\Delta\omega\tau = 2\pi N$$
$$\tau = \frac{2(l_1 - l_2)}{c}$$
$$\Delta x = \frac{N\pi c}{\Delta\omega}$$

Numerically, the first crossing is when

$$\Delta x = \frac{\pi c}{\Delta\omega} = 0.01142 \text{ m}$$

2. Realistic Sodium Doublet Theory (12 pts) In this question, you'll calculate and study the features of Michelson interference fringes for a more realistic model of input light from a low-pressure sodium discharge lamp. Most of the setup from Problem 1 still applies, except:

1. Each of the two atomic transitions responsible for the sodium doublet emission has natural linewidth of $2\pi(9.8\text{MHz})$. However, the operating temperature of sodium vapor in a typical low-pressure lamp is 260°C, so Doppler broadening completely swamps the natural linewidths. What is the standard deviation σ_ω of these Doppler-broadened sodium emission lines? Technically the two standard deviations are very slightly different from each other, but neglect the difference! Calculate the ratio $\frac{\sigma_\omega}{\Delta\omega}$, where $\Delta\omega$ is the difference between the angular frequencies of the two lines (as defined in Problem 1).

$$s(\omega) = \frac{1}{\sqrt{2\pi\sigma_\omega{}^2}} e^{-(\omega-\omega_0)^2/2\sigma_\omega{}^2}$$
$$\sigma_\omega{}^2 = \left(\frac{\omega_0}{c}\right)^2 \frac{k_B T}{m}$$

We can calculate this.

We know that

$$\Delta\omega = 8.244 \times 10^{10}$$
$$\omega_0 = 8.097 \times 10^{13}$$

the mass of sodium is $m = 22.99$ grams per mol

This gives us

$$\sigma_w^2 = 1.792854611 \times 10^{-10}$$
$$\sigma_w = 0.0000133915950597$$

This gives us

$$\frac{\sigma_\omega}{\Delta\omega} \approx 0.00143$$

2. Assuming the two transitions emit with equal intensities, write the normalized spectral density $s(\omega)$ for the sodium doublet emission. Recall that $s(\omega)$ is normalized so that $\int_0^\infty s(\omega)d\omega = 1$.

$$s(\omega) = \frac{1}{\sqrt{2\pi\sigma_\omega{}^2}} e^{-(\omega-\omega_0)^2/2\sigma_\omega{}^2}$$
$$\sigma_\omega{}^2 = \left(\frac{\omega_0}{c}\right)^2 \frac{k_B T}{m}$$

$$1 = \frac{1}{2}\frac{1}{\sqrt{2\pi\sigma_\omega^2}}\int_0^\infty e^{\frac{-(\omega-\omega_0)^2}{2\sigma_\omega^2}} d\omega + \frac{1}{2}\frac{1}{\sqrt{2\pi\sigma_\omega^2}}\int_0^\infty e^{\frac{-(\omega-\omega_1)^2}{2\sigma_\omega^2}}$$

I calculated this to check that it was normalized (it was), then deleted it since I thought I did the wrong thing and started on the work below. Because we aren't asked to explicitly check, I figure it is fine to just leave the formula.

old work

I am confused by this - the next step is calculating the coherence from the Doppler shifting intensity function, so I can't use the coherence function from before. But that means this is just copying from the slides. I started the derivation if we were using the $g^{(1)}$ that we calculated in problem 1 for two different equal intensity frequencies.

$$Re(g^{(1)}(\tau)) = \frac{E_1^{(+)}E_1^{(-)}\cos(\omega_1\tau) + E_2^{(+)}E_2^{(-)}\cos(\omega_2\tau)}{E_1^2 + E_2^2}$$

If we have equal amplitudes, this is

$$Re(g^{(1)}(\tau)) = \frac{E_0^2(\cos(\omega_1\tau) + \cos(\omega_2\tau))}{2E_0^2}$$
$$= \frac{1}{2}(\cos(\omega_1\tau) + \cos(\omega_2\tau))$$

$$s(\omega) = \frac{2}{\pi}\int_0^\infty \mathrm{Re}\left[g^{(1)}(\tau)\right] \cos\omega\tau \, d\tau$$

We therefore have

$$\frac{1}{\pi}\int_0^\infty \cos(\omega_1\tau)\cos(\omega\tau) + \cos(\omega_2\tau)\cos\omega\tau \; d\tau$$

We can rewrite this using trig identities

Lets use a dummy variable x, since this process will look the same for ω_1 and ω_2 in the overall integral.

$$\frac{1}{\pi}\int_0^\infty \cos(x\tau)\cos(\omega\tau)d\tau$$
$$= \frac{1}{2\pi}\int_0^\infty \cos((x+\omega)\tau) + \cos((x-\omega)\tau) \; d\tau$$

This doesn't converge. What.

$$\frac{1}{\pi}\int_0^\infty 2\cos\left(\frac{\omega_1+\omega_2}{2}\right)\cos\left(\frac{\omega_1-\omega_2}{2}\right)\cos\omega\tau \, d\tau$$

3. What is the coherence function $g^{(1)}(\tau)$ for the sodium doublet? Hint: After relating $g^{(1)}(\tau)$ to $s(\omega)$ via an integral, extend that integral to negative values of ω without changing the result, since $s(\omega) \approx 0$ far from ω_1 and ω_2. Now many references will tell you how to evaluate the integral; see, for example, App. D of the Ph 116 text.

Using the formula

$$g^{(1)}(\tau) = \int_0^\infty S(\omega) e^{-i\omega\tau} d\omega$$
$$\frac{1}{\sqrt{2\pi\sigma_\omega^2}} \int_0^\infty e^{\frac{-(\omega-\omega_0)^2}{2\sigma_\omega^2}} e^{-i\omega\tau} d\omega$$

I will do this once, using ω_0. This will be the exact same result as the second part of the gaussian, which will replace ω_0 with ω_1.

Lets relabel $\omega - \omega_0$ as x. This is symmetric about 0, so we can make the integral easier by extending the bounds. Lets also relabel $a = \frac{1}{2\sigma_\omega^2}$

$$\begin{aligned}
&\frac{1}{\sqrt{2\pi\sigma_\omega^2}} \int_{-\infty}^\infty e^{-ax^2} e^{-i(x+\omega_0)\tau} dx \\
&= \frac{1}{\sqrt{2\pi\sigma_\omega^2}} \int_{-\infty}^\infty e^{-ax^2 - i\tau x - i\omega_0\tau} dx \\
\text{let } b \equiv i\tau \quad & \\
&= \frac{1}{\sqrt{2\pi\sigma_\omega^2}} e^{-i\omega_0\tau} \int_{-\infty}^\infty e^{-(ax^2+bx)} dx
\end{aligned}$$

Following the derivation of Appendex D of Townsend "A modern approach to quantum mechanics", this is in the same form as EQ D.5 with the result D.7

$$\int_{-\infty}^\infty dx e^{-ax^2+bx} = e^{\frac{b^2}{4a}} \sqrt{\frac{\pi}{a}}$$

Our problem is now
$b^2 = -\tau^2$. $4a = \frac{2}{\sigma_\omega^2}$

$$\begin{aligned}
g^{(1)}(\tau)_{\omega_0} &= \frac{1}{\sqrt{2\pi\sigma_w^2}} e^{-i\omega_0\tau} e^{\frac{-\tau^2\sigma_\omega^2}{2}} \sqrt{2\pi\sigma_w^2} \\
&= e^{-i\omega_0\tau} e^{-\frac{\tau^2\sigma_\omega^2}{2}}
\end{aligned}$$

We have to add the component from the other frequency of the doublet, so this becomes

$$\begin{aligned}
g^{(1)}(\tau) &= \frac{1}{2} \frac{1}{\sqrt{2\pi\sigma_w^2}} e^{-i\omega_0\tau} e^{\frac{-\tau^2\sigma_\omega^2}{2}} \sqrt{2\pi\sigma_w^2} \\
&= \frac{1}{2} e^{-\frac{\tau^2\sigma_\omega^2}{2}} (e^{-i\omega_0\tau} + e^{-i\omega_1\tau})
\end{aligned}$$

4. Calculate the coherence time $\tau_{coh} = \int_{-\infty}^{\infty} \left|\mathrm{Re}\left[g^{(1)}(\tau)\right]\right|^2 d\tau$ and the coherence length $\ell_{coh} = c\tau_{coh}$. Make a reasonable approximation, based on the value of $\frac{\sigma_\omega}{\Delta\omega}$, to help you evaluate the necessary integral here!

$$g^{(1)}(\tau) = \frac{1}{2} e^{-i\omega_0\tau} e^{-\frac{\tau^2\sigma_\omega^2}{2}}$$

Therefore, we have

$$\frac{1}{2} \int_{-\infty}^\infty \left| \cos(-\omega_0\tau) e^{\frac{-\tau^2\sigma_\omega^2}{2}} \right|^2 d\tau$$

We can rewrite this and combine.

$$\cos(x) = \frac{e^{ix} + e^{-ix}}{2}$$

This gets

$$\frac{1}{4}\int_{-\infty}^{\infty}\left|e^{i\omega_0\tau-\tau^2\sigma_\omega^2}+e^{-i\omega_0\tau-\tau^2\sigma_\omega^2}\right|^2 d\tau$$
$$=\frac{1}{4}\int_{-\infty}^{\infty} 2e^{-2\tau^2\sigma_\omega^2}+e^{2i\omega_0\tau-2\tau^2\sigma_\omega^2}+e^{-2i\omega_0\tau-2\tau^2\sigma_\omega^2}\ d\tau$$

Lets break this into three Gaussians

Gaussian 1

$$2e^{-2\tau^2\sigma_\omega^2}$$

These have well explored solutions.

$$\int_{-\infty}^{\infty} e^{-ax^2}=\sqrt{\frac{\pi}{a}}$$

This whole expression evaluates to

$$\frac{1}{2}\sqrt{\frac{\pi}{2\sigma_\omega^2}}$$

Gaussian 2

$$e^{2i\omega_0\tau-2\tau^2\sigma_\omega^2}$$
$$\int_{-\infty}^{\infty} e^{-ax^2+bx}=e^{\frac{b^2}{4a}}\sqrt{\frac{\pi}{a}}$$

Here, $b^2=-4\omega_0^2$
and $\frac{1}{4a}=\frac{1}{8\sigma_\omega^2}$
This whole expression gives us

$$\frac{1}{4}e^{\frac{-\omega_0^2}{2\sigma_\omega^2}}\sqrt{\frac{\pi}{2\sigma_\omega^2}}$$

Gaussian 3

$$e^{-2i\omega_0\tau-2\tau^2\sigma_\omega^2}\ d\tau$$

$b^2=-4i\omega_0^2$.
$\frac{1}{4a}=\frac{1}{8\sigma_\omega^2}$
We have the same process as for Gaussian 2. This gets the same exact result as Gaussian 2.

Continuing the result

We now have for ω_0:

$$\tau_{\text{coh}}=\frac{1}{2}\sqrt{\frac{\pi}{2\sigma_\omega^2}}\left(1+e^{\frac{-\omega_0^2}{2\sigma_\omega^2}}\right)+\frac{1}{2}\sqrt{\frac{\pi}{2\sigma_\omega^2}}\left(1+e^{\frac{-\omega_1^2}{2\sigma_\omega^2}}\right)$$

And the coherence length

$$\ell_{\text{coh}}=\frac{c}{2}\sqrt{\frac{\pi}{2\sigma_\omega^2}}\left(1+e^{\frac{-\omega_0^2}{2\sigma_\omega^2}}\right)+\frac{c}{2}\sqrt{\frac{\pi}{2\sigma_\omega^2}}\left(1+e^{\frac{-\omega_1^2}{2\sigma_\omega^2}}\right)$$

This gives

$$\tau_{\text{coh}}=9.3*10^{-6}\ \text{seconds}$$
$$\lambda_{\text{coh}}=2808.07\ \text{m}$$

old work

$$= \frac{1}{2}\int_{-\infty}^{\infty} 2e^{2\tau^2\sigma_\omega^2} + e^{(2\tau^2\sigma_\omega^2)}(e^{2i\omega_0\tau} + e^{-ti\omega_0\tau})d\tau$$
$$= \frac{1}{2}\int_{-\infty}^{\infty} 2e^{2\tau^2\sigma_\omega^2} + 2e^{(2\tau^2\sigma_\omega^2)}\cos(2\omega_0\tau)\,d\tau$$
$$= \frac{1}{2}\int_{-\infty}^{\infty} 2e^{2\tau^2\sigma_\omega^2}(1+\cos(2\omega_0\tau))\,d\tau$$

5. Consider placing this lamp at the input of a Michelson interferometer. Give a formula for the normalized output intensity $\frac{I_{OUT}(\tau)}{I_{IN}}$ that should be observed. Sketch a (probably not to scale) plot of the output intensity as a function of τ, noting important features and axis scales.

$$g^{(1)}(\tau) = \frac{1}{2}e^{-i\omega_0\tau}e^{-\frac{\tau^2\sigma_\omega^2}{2}}$$

We know that

$$I_{\text{out}} = \frac{1}{2}I_{in}[1 - Reg^{(1)}(t)]$$

This gives the relative intensity

$$\frac{I_{\text{out}}}{I_{\text{in}}} = \frac{1}{2}\left(1 - \frac{1}{2}\cos(\omega_0\tau)e^{\frac{-\tau^2\sigma_\omega^2}{2}}\right)$$

6. What distance $\delta\ell_{sep}$ must the interferometer's translatable mirror be moved in order to resolve the separation between the two sodium lines (*i.e.*, to get visibilty vs. τ data you can use to find the separation between the two sodium lines)? What distance $\delta\ell_{width}$ must it be moved in order to resolve the Doppler-broadened linewidth of the individual lines? Clearly explain how you arrive at these answers, and also evaluate your answers as numerical distances.

However, the bands are Doppler broadened - but we really care about the distance between the centers of the two Gaussians that describe the light. That is the distance that one must travel to see light patterns from alignment of one frequency and not the other.

If we thought that the beams were perfect (no broadening), then we would see peaks of them offset by a length corresponding the difference in the frequencies. The seperation in frequency between them is

$$2\Delta\omega = 2 * 8.244 \times 10^{10}$$

We can convert this to length as

$$\begin{aligned} 2\pi\omega\lambda &= c \\ \delta\ell_{\text{ width}} &= \frac{c}{2\pi\omega} \\ \delta\ell_{\text{ width}} &= \frac{c}{4\pi\Delta\omega} \\ &= 0.000289m \end{aligned}$$